Vishnu Sakthi Selvakumar

Vishnusakthi.me | vishnus1793 | Vishnu S | vishnusakthi.44@gmail.com | 7397286386

PROFESSIONAL SUMMARY

I aim to apply my technical skills to build practical solutions and contribute effectively to a team, while continuously learning and improving in a professional environment.

PROJECTS

ROOM BOOKING SYSTEM | Flask, MongoDB, JWT

- ➔ Built full-stack booking platform with user/admin roles.
- ➔ Implemented JWT authentication and secure password hashing (bcrypt).
- ➔ Developed booking system with conflict-free date validation.
- ➔ Designed REST APIs and managed MongoDB collections efficiently.
- ➔ Created admin panel for booking approval and room management.

WEB SCRAPER & MIND MAP GENERATOR | FastAPI, BeautifulSoup, Sumy

- ➔ Built a FastAPI backend to scrape and structure web content.
- ➔ Used LSA and TextRank with AI fallback for summarization.
- ➔ Converted raw data into meaningful summaries.
- ➔ Generated hierarchical and network-based mind maps.
- ➔ Stored outputs as optimized JSON for efficiency.

MULTI-SEARCH ENGINE SUMMARIZER | Python, LLaMA3, Ollama, AsyncIO

- ➔ Integrated Google and DuckDuckGo search with asynchronous execution.
- ➔ Reduced redundant API calls using 24-hour caching mechanism.
- ➔ Enabled local summarization using LLaMA3 via Ollama and LLaMAIndex.
- ➔ Implemented parallel scraping with robust error handling and deduplication.
- ➔ Designed modular pipeline for search, processing, and summarization.

STOCK PRICE PREDICTION USING MACHINE LEARNING | Python, Scikit-learn, Time-Series Forecasting

- ➔ Built a machine learning pipeline using historical financial data from Yahoo Finance.
- ➔ Engineered time-series features including lag (previous close), and calendar-based features (day, month, year).
- ➔ Implemented Linear Regression using Scikit-learn for stock price prediction.
- ➔ Performed train-test split for model validation and generalization.
- ➔ Evaluated model performance using RMSE (Root Mean Squared Error).
- ➔ Generated real-time predictions using the latest available market data.

ACHIEVEMENTS

2ND PLACE – 30-HOUR HACKATHON | KEC

- ➔ Built full-stack room booking platform with 360° visualization.
- ➔ Delivered functional prototype within 30 hours.
- ➔ Collaborated in a 3-member team following rapid Agile workflow.

3RD PLACE – 24-HOUR HACKATHON | KEC

- ➔ Developed high-performance web scraping engine.
- ➔ Structured and cleaned large-scale product data efficiently.
- ➔ Optimized execution speed under strict 24-hour constraints.

EDUCATION

KONGU ENGINEERING COLLEGE, PERUNDURAI, ERODE

B.Tech in Artificial Intelligence and Machine Learning

2022 - 2026

CGPA: 7.33 / 10.0

KONGU VELLALAR MATRIC HIGHER SECONDARY SCHOOL, ERODE

Higher Secondary Certificate (HSC)

79%

NANDHA MATRIC HIGHER SECONDARY SCHOOL, ERODE

Secondary School Leaving Certificate (SSLC)

69%

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL

Frontend Technologies: React.js, HTML5, CSS3, JavaScript

Backend Technologies: Flask, FastAPI, Node.js, REST APIs

Databases: MongoDB, SQL, Database Management Systems (DBMS)

DevOps & Cloud: Docker, CI/CD, Terraform, AWS (EC2, S3), Linux

CERTIFICATIONS

- GitHub Foundations
- MongoDB Associate Developer
- Docker Mastery: Kubernetes and Swarm